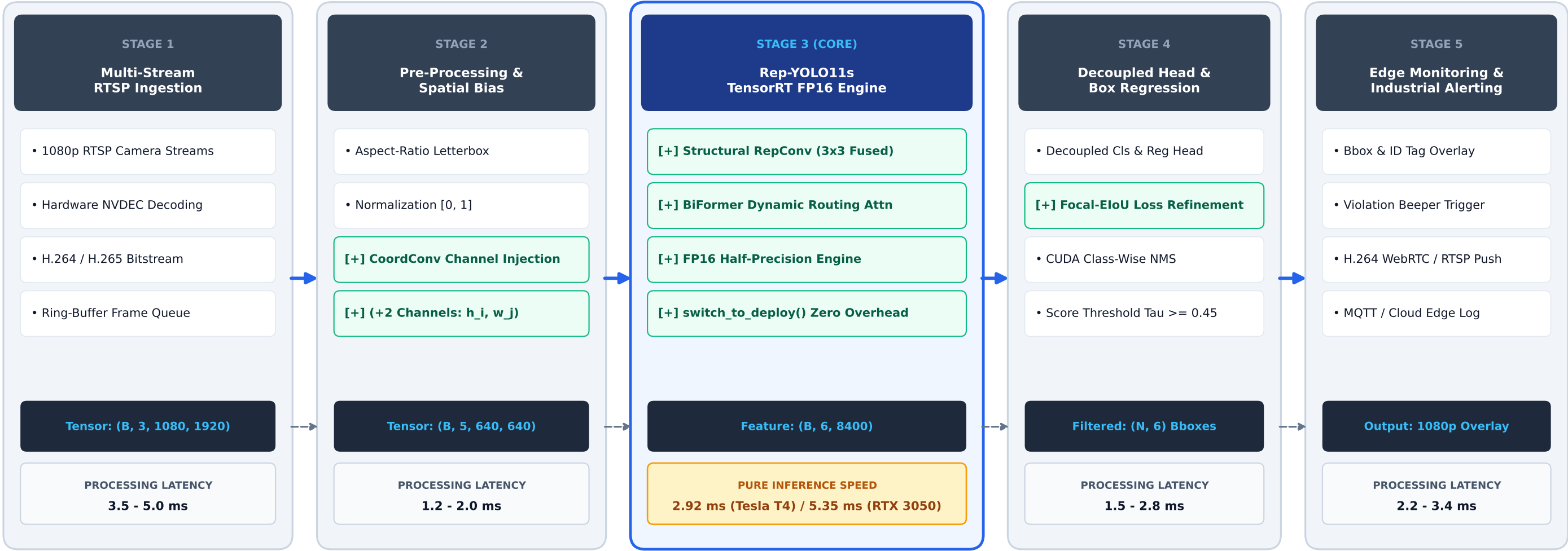


END-TO-END INDUSTRIAL PPE COMPLIANCE & SAFETY HELMET SURVEILLANCE PIPELINE

Real-Time Multi-Stream RTSP Ingestion • CoordConv Spatial Bias • Structural RepConv • TensorRT FP16 Acceleration (342.5 FPS)



LEGEND & INNOVATION HIGHLIGHTS:

[*] Proposed Architectural Innovations

[!] GPU Throughput: 342.5 FPS (T4) / 187.1 FPS (RTX 3050)

Multi-Camera Pipeline: 65 - 95 FPS (10.5 - 15.4 ms E2E)